Module MOD1
Geoinformation Systems (GIS) Application

DETERMINATION OF THE NORMALIZED DIFFERENCE WATER INDEX (NDWI) FOR THE LAKE CHAD BASIN

Project 11

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ABSTRACT

Lake Chad which was among the largest lakes in Africa has experienced about 90% reduction of its water level since 1963. The major contributions of Lake Chad are its support for urbanization and serving the needs of millions of people across the countries it borders (Chad, Nigeria, Cameroon, and the Niger Republic). In this study, surface water changes in Lake Chad were mapped. These changes in surface water are constantly and frequently updated via the remote sensing technology. We considered the effect of drought by choosing Landsat images of the same range in period for two different years (2015 and 2019). The Normalized Difference Water Index (NDWI) was used to investigate surface water bodies in the basin. Also, for the purpose of accuracy, another index known as the Modified Normalized Difference Water Index (MNDWI) was used. In addition, the NDWI_V which is the water index for vegetation was also included to check for the moisture content of plant leaves in the study area. Obtained results display the effectiveness of NDWI, MNDWI, and NDWI_V techniques in mapping surface water and Geographic Information System for the detection of changes in surface water at different times.

Keywords: Urbanization; Landsat; NDWI; MNDWI; NDWI_V; remote sensing

INTRODUCTION

An important feature of earth's ecosystems is surface water. It plays an integral role in the survival of humans (Vörösmarty et al., 2000), and used to determine changes in the environment (Pekel et al., 2016). Surface water inventory mapping, water estimation for drinking and irrigation purposes, land use, and land cover are some of the scientific works that solely depend on updated and accurate information on the distribution of surface water in space. According to Pekel et al (2016), surface water patterns are altered by anthropogenic activities, but there is also a response to human behavior due to surface water distribution.

However, these alterations in surface water patterns have elicited serious questions and concerns about water security and its management across borders (Ashtekar, 2019). Also, alternate wet and dry seasons in a year including a drift in location can indicate significant changes in seasonal water surfaces. All in all, Downing et al (2006), stated that the changing surface water has a far-reaching impact on humans and biodiversity, but the improvement of water resource management and in-depth knowledge of the connectivity, time characteristics, and the consequences of land management decisions depend on precise mapping. One important surface water mapping technique is the normalized difference water index (NDWI). Zhou et al (2017) stated that the NDWI with the aid of Landsat images can be utilized to accurately map water bodies.

The most fundamental NDWI was developed by Mcfeeters (1996), using green and near-infrared (NIR) bands of a Landsat thematic mapper (TM) to maximize the identification of water features. NDWI values range from -1 to 1 and he used a threshold of 0. If NDWI is > 0 an area is said to be occupied by water while a value ≤ 0 indicates the absence of water. Our aim for this study is to use Landsat images obtained from the Lake Chad Basin to calculate the normalized difference water index. The NDWI values and observed patterns will give us information on the Spatio-temporal changes in the water levels at this basin.

Besides the NDWI, we also used the Modified Normalized Difference Water Index (MNDWI) for the identification of water bodies, which was introduced by Xu (2006). Here, the short-wave infrared (SWIR) band is used instead of NIR. The advantage of the MNDWI is the ability to enhance open surface water features while suppressing built-up land, vegetation and soil noise. In addition to the investigation of water bodies, we examine the

change of the water content of the vegetation. This is done by following the approach of Gao (1996), using the NIR and the SWIR band. Since this index is also called NDWI, we will use the expression NDWI_V for this index to avoid confusion.

STUDY AREA

The Lake Chad Basin consists of an interconnected water network which includes Lake Chad; it covers Cameroon, Niger, Nigeria, Chad and the Central African Republic, providing natural resources to the region for over 30 million people (Leblanc et al., 2007). Lake Chad is located at an altitude of 282m above sea level in the Sahel region of Central Africa where the Sahara Desert meets the Savannah lands and its catchment area is 2.53 x 10^6 km^2 large (FEWS, 1997) (figure 1). The climate around the lake is semiarid in the south and arid in the north with a dry season from October to April, and a short rainy season from May to August (usually 550mm annually in the south and 250mm annually in the north) (Birkett, 2000).

Lake Chad is defined as "closed" referring to zero water outflow, though minor outflows along the Bahr el Ghazal occurred prior to 1875 (ILEC, 1988; Hutchinson et al., 1992). With low humidity, high temperatures and shallow depths of the lake ranging 1.5–8 m, most of the lake's water is lost by evaporation (Roche, 1973) with related annual water level fluctuations. However, according to Buma et al. (2018), the key causes of changes in the lake level were climate variability and environmental degradation. In their study, Coe & Foley (2001) show that between 1983 and 1994, water used for irrigation in the Lake Chad region increased fourfold than between 1953 and 1979. Another research used a hydrological model to predict the impact of bathymetry, anthropogenic activities and climate variability on lake level, surface area, and water storage (Gao et al., 2011). They discovered that siltation from the lake and extreme droughts caused the lake to split in 1972. Since then, the structure of the lake chad basin is characterised by a northern basin and a southern basin, divided by a vegetation-covered sand barrier called the “Great Barrier” (see figure 1). The authors have also stressed irrigation as the key cause of water depletion though they find it challenging to evaluate the exact amount of existing water within the Lake Chad Basin.

According to Buma et al. (2018), Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) has been used to show water loss/fluctuation within the Chad basin where the lake shoreline recedes. Numerous studies have been conducted with the goal of understanding these fluctuations and offering some form of solution for improving the situation in Lake Chad (Coe & Foley, 2001; Luxereau et al., 2011).

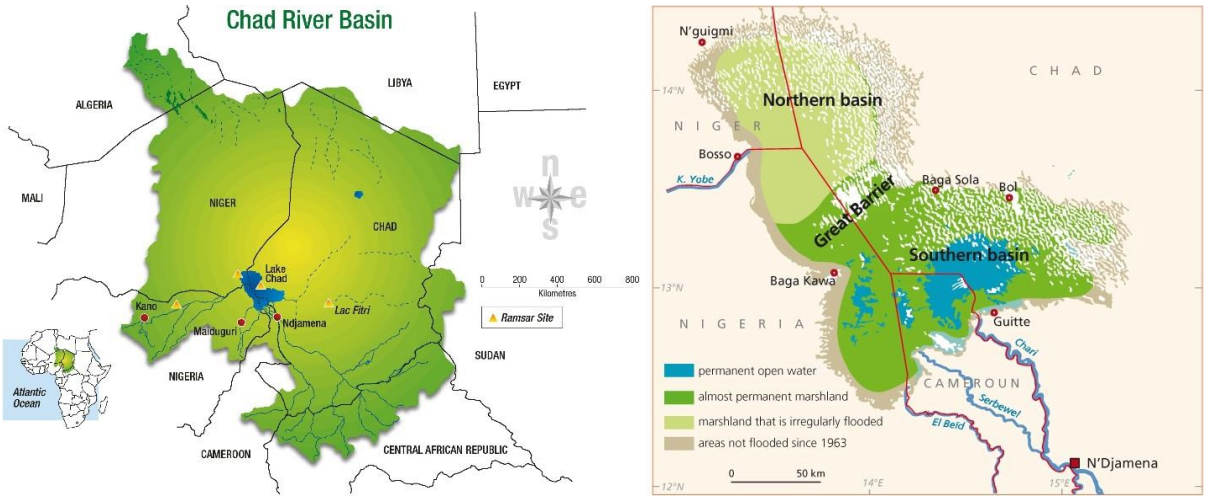

Fig 1: Map of study area. Sources : *wwf.panda.org* and *africa-water-environmentalchange.blogspot.com*

DATA AND METHODOLOGY

For this study, we downloaded two Landsat images with 30m resolution of the collection 1, level -1, and Operational Land Images/Thermal Infrared sensors (OLI/TIRS) Landsat 8 from Earth Explorer (www.earthexplorer.usgs.gov), an archive system owned by the United States Geological Survey (USGS). One Landsat scene with a path/row number of 185/51 was used to generate the total coverage of the study area because it consists of about 80% of the spatial extent of Lake Chad. We also considered the effect of drought by choosing images of the same time range for two different years (01/05/2015 – 30/09/2015 and 01/05/2019 – 30/09/2019). The reason was that we intend to compare our maps for the selected times in order to observe changes in patterns and also to determine the changes in water level or the availability of water in the lake over time. The acquired data were pre-georeferenced to UTM zone 33 north projection with WGS-84 datum. Finally, we used the R 3.6.2 software package to process and visualize our data. The equations for the three indices are as follows:

$$NDWI = \frac{Green - NIR}{Green + NIR}$$

$$MNDWI = \frac{Green - SWIR}{Green + SWIR}$$

$$NDWI_V = \frac{NIR - SWIR}{NIR + SWIR}$$

For Green, NIR and SWIR we used the Landsat 8 bands 3, 5 and 6, respectively.

RESULTS

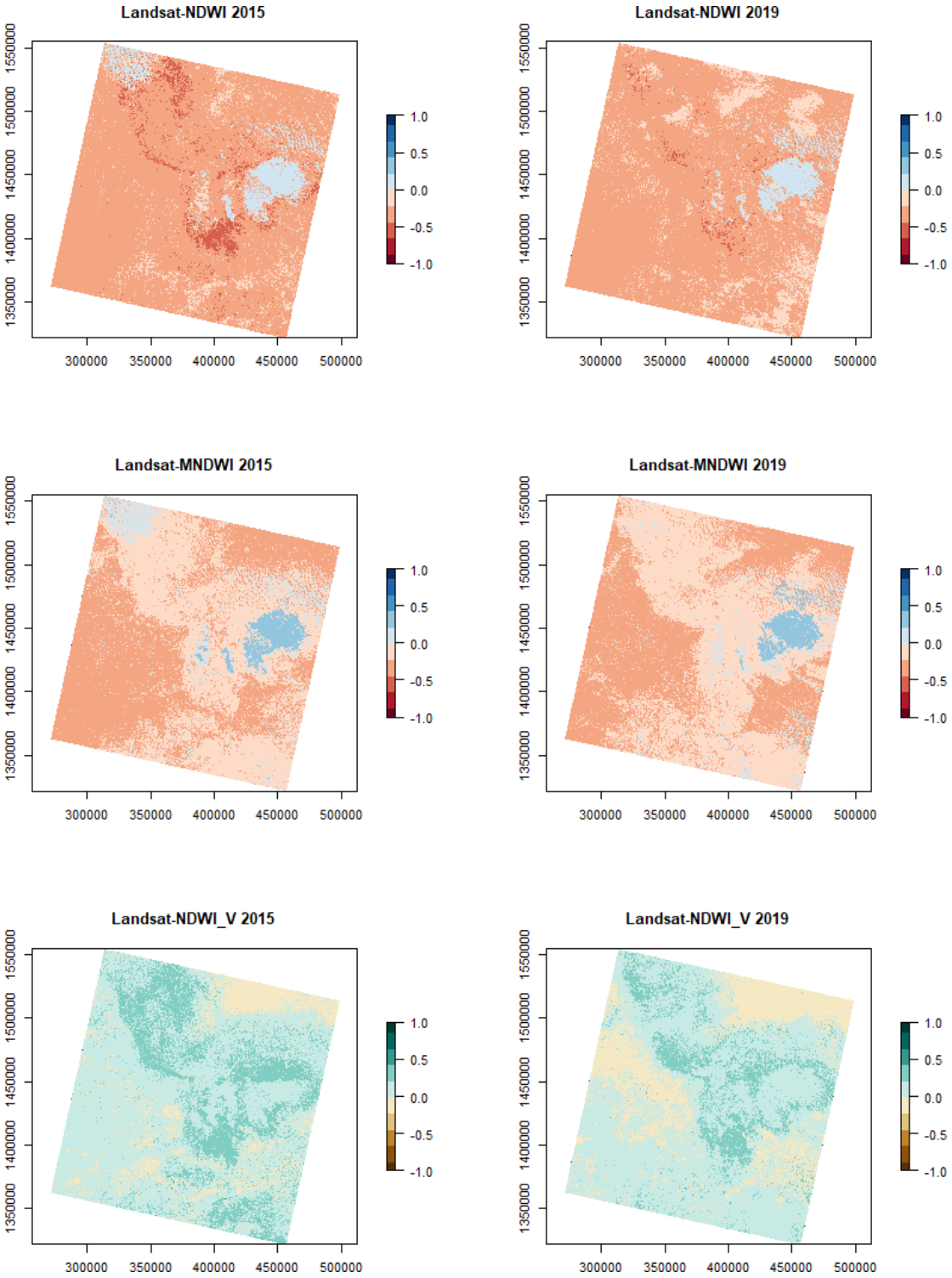

Figure 2: Results of the NDWI, MNDWI and NDWI_V for the years 2015 and 2019

This segment refers to the results of figure 2. All three water indices show a difference when comparing the years 2015 and 2019. A difference well visible is at the northern basin (top left corner). In this region, the NDWI and the MNDWI identified scattered water bodies in 2015. In 2019 however, these water bodies are missing or they are decreased to a great degree. In contrast, the changes of the water bodies in the southern basin (right side) are very minor. Here, the large and connected water body doesn't show a considerable change in its area. The area of the water bodies south-east from that indicates a slight decline, while the scattered water bodies further north indicate a slight increase. When comparing the NDWI_V of 2015 and 2019 two major trends are visible. Firstly, the light brown areas (values below 0) increase, well visible on the left side at the middle. Secondly, a decrease of the darker green areas (values from ~0.2 to 0.4) are detectable at the top left corner, the bottom right corner and in the north of the large water body in the southern basin. Besides a temporal comparison it's also possible to compare the indices of the NDWI and the MNDWI. Overall the NDWI seems to produce lower values compared to the MNDWI, which is noticeable in the light red areas (values from ~ -0.2 to 0) and the darker red areas (values < -0.2). The NDWI shows larger areas with darker red, while the image of the MNDWI is characterised by larger light red areas. The light red areas of the MNDWI seem to fit the area of the marshlands quite well (compare figure 1). Compared to that, the NDWI doesn't show this contrast of the marshlands to the dryer areas outside of the basin. Another aspect are the darker blue areas (values > 0.2) of the MNDWI representing (for example) the large water body in the southern basin, while the NDWI doesn't produce darker blue areas at all.

DISCUSSION

A key outcome of our study is the aspect that the northern basin is affected by large changes in its water area, while the southern basin seems to be far more stable. This fits the findings of Pham-Duc et al. (2020), who show that the surface water extent of the northern basin is characterised by high fluctuations in contrast to minor fluctuations of the southern basin. Furthermore they show that in 2015 the surface water extent in the northern basin was unusually high, which also fits our image. Surface water extents significantly smaller than in 2015 are the norm in the last two decades. According to this, our image of 2019 is no surprise. It seems that the main difference between 2015 and 2019 can be explained by the aspect that the year 2015 was more humid compared to 2019. This aspect is supported by the image of the NDWI_V. The difference between the images is clearly in the range of usual fluctuations between the years. If we would have chosen the year 2012 instead of 2015, a comparison with 2019 probably wouldn't show such a visible difference (considering the data of Pham-Duc et al., 2020).

Since we only compare images of two points in time and considering the strong fluctuations of the water area in the northern basin, extrapolating a trend based on our data would be connected with a high probability of error. For example, if we would have compared the years 2012 and 2015 the result would be an increasing trend of the water area, while our data show a drastic decrease. In this case a meaningful statement about a trend can only be made with the support of other studies. This is true for the northern basin. The southern basin however shows way finer fluctuations and extrapolating a trend would be less connected with errors. Here our study shows a steady trend for the water area, which is in line with the data of Pham-Duc et al. (2020). Unsurprisingly the drastic decreasing trend of the northern basin is not in line with that study, which only shows a very slow decrease of the surface water extent over two decades.

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